MS ENGINEERING COLLEGE, BANGALORE

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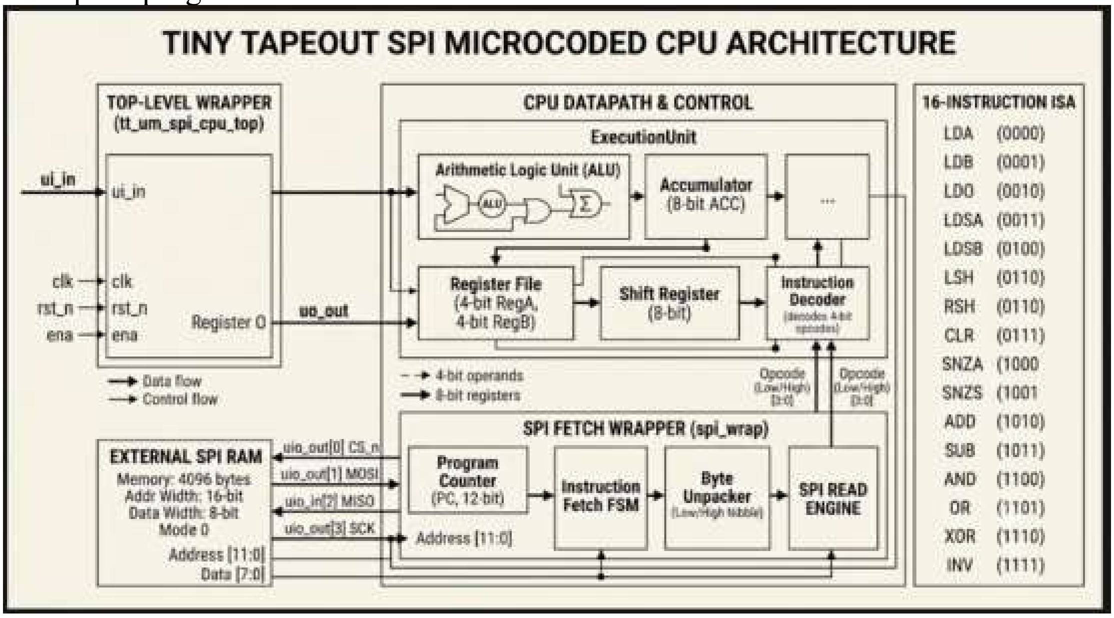
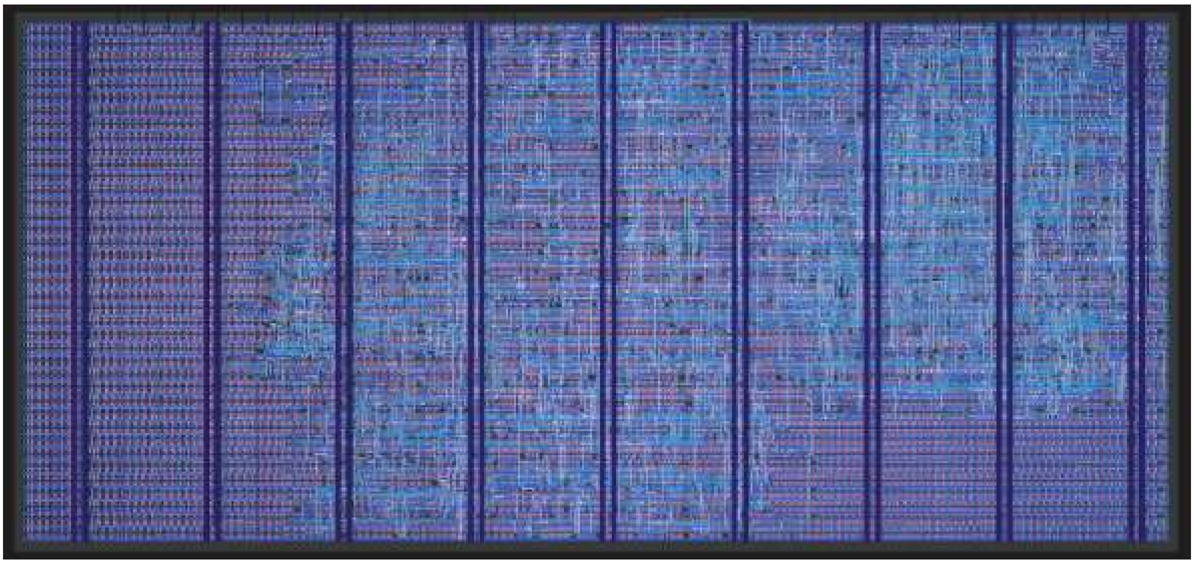
Chip Tapeout Project Details

SPI MICROCODED CPU

The tt_spi_microcoded_cpu is a compact 4-bit microcoded CPU specifically designed for the TinyTapeout (GF180MCU) platform. Unlike traditional embedded processors, it saves space by fetching its 16-instruction ISA directly from an external SPI RAM using a unique dual-instruction-per-byte encoding. The architecture features a custom execution unit with a dedicated ALU, accumulator, and shift registers. While it contains a couple of known hardware limitations—such as non-functional conditional skip instructions (SNZA/SNZS) and the lack of a HALT mechanism—it successfully demonstrates hardware functionality using a fully verified shift-and-add 4×4-bit multiplier program.



Block Diagram of SPI MICROCODED CPU



GDSII of the Design SPI MICROCODED CPU

THE TEAM AND THEIR CONTRIBUTIONS

A dedicated team of students has successfully designed and implemented an **SPI Microcoded CPU**, showcasing their growing expertise in digital system design, computer architecture, and semiconductor engineering. The project commenced with an in-depth study of the Serial Peripheral Interface (SPI) protocol and microcoded processor architecture, followed by the development of the CPU data path, control unit, instruction sequencing, and SPI communication interface using Verilog HDL. Extensive functional verification and simulation were carried out to validate instruction execution, SPI data transfers, and overall processor functionality while optimizing the design for performance and resource utilization. To realize the design as a manufacturable integrated circuit, the students adopted the **OpenLane** open-source ASIC design flow. The RTL design was synthesized into a gate-level netlist, followed by floorplanning, power planning, standard cell placement, clock tree synthesis, routing, static timing analysis, Design Rule Checking (DRC), and Layout Versus Schematic (LVS) verification, ultimately producing a fabrication-ready GDSII layout. Through this end-to-end implementation, the students gained practical exposure to the complete RTL-to-GDSII ASIC design methodology employed in the semiconductor industry. The project provided invaluable hands-on experience in processor design, digital communication interfaces, hardware verification, and physical IC implementation, effectively bridging the gap between theoretical learning and real-world semiconductor product development. It reflects the institution's commitment to nurturing industry-ready VLSI engineers equipped with practical expertise in open-source Electronic Design Automation (EDA) tools, modern ASIC design methodologies, and System-on-Chip (SoC) development.

The TEAM (in include all names)		
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CHIP TAPEOUT PROGRAM

Module / duration	1 (April)	2 (May)	3 (Nov.)	4 (Nov.)	5 (Dec.)	6 (Dec.)	7
Fast track chip design	✓ Completed						
Chip tapeout submission Global Foundry 180nm		✓ 11-05-2026 Completed					
Hands on experience for familiarisation		✓	✓	✓	✓		
Chip fabrication			✓ 04-11-2026				

			In process				
Packaging				✓ 04-11-2026 Pending			
Chip shipment					20-12-2026 Pending		
Chip testing						20-12-2026 Pending	
Product testing							3C0-12-2026 Pending

HOW IS THIS PROJECT RELEVANT AND IMPORTANT FOR THE STUDENTS

The **SPI Microcoded CPU** project provides students with a comprehensive understanding of processor architecture, digital system design, and semiconductor implementation. Developing a microcoded processor with an SPI communication interface enables students to acquire practical knowledge in:

- Processor architecture and datapath design
- Microcoded control unit development
- Instruction set implementation and execution
- Finite State Machine (FSM) design
- Register file and Arithmetic Logic Unit (ALU) design
- Serial Peripheral Interface (SPI) protocol implementation
- Clock generation and synchronization
- RTL coding using Verilog HDL
- Functional verification and simulation
- ASIC implementation using the OpenLane RTL-to-GDSII design flow

These competencies form the foundation for careers in digital IC design, processor development, FPGA design, ASIC design, embedded systems, and System-on-Chip (SoC) engineering.

SCOPE FOR FUTURE ENHANCEMENT

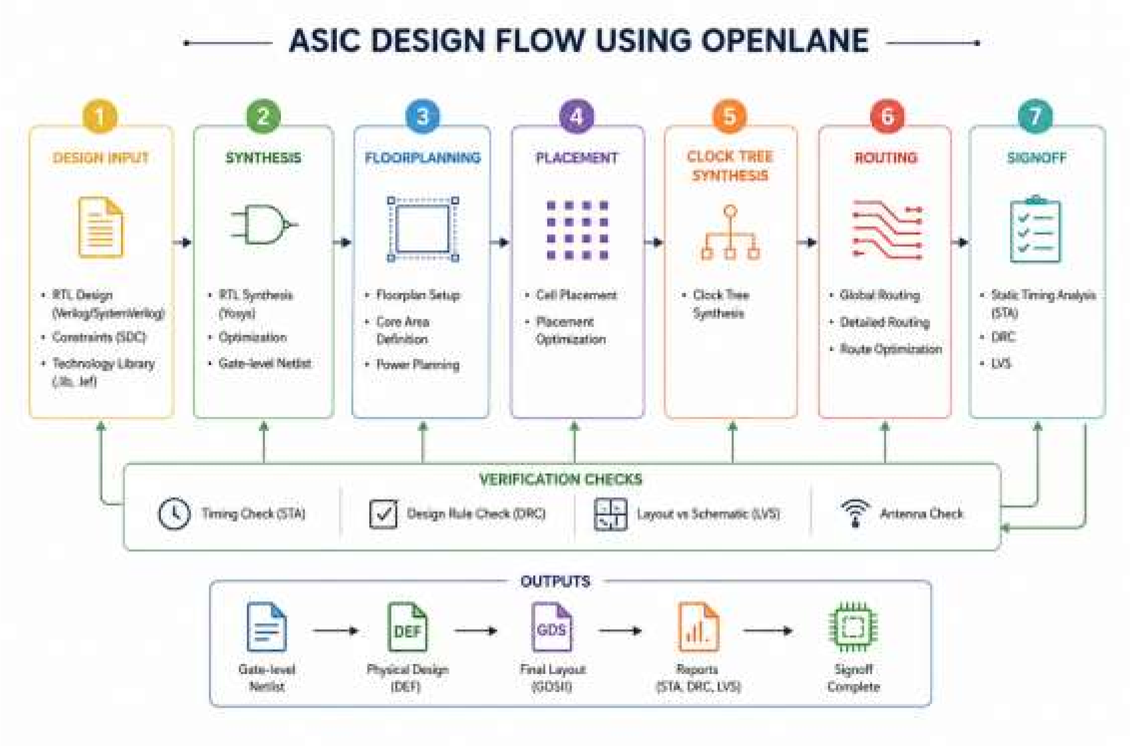
The project provides an excellent platform for advanced processor and SoC development. Students can further enhance the design by:

- Expanding the instruction set architecture (ISA)
- Implementing interrupt and exception handling mechanisms
- Adding pipelining for improved processor performance
- Integrating instruction and data cache memories
- Supporting external memory interfaces (SRAM/DDR)
- Integrating the SPI Microcoded CPU into a RISC-V or custom SoC platform
- Developing a UVM-based verification environment for comprehensive functional verification
- Implementing low-power optimization techniques such as clock gating and power gating
- Comparing FPGA implementation with ASIC realization for performance, power, and area analysis

- Adding additional communication interfaces such as UART, I²C, CAN, or SPI Slave functionality
- Incorporating hardware security features such as secure boot and cryptographic accelerators
- Optimizing the processor for higher operating frequency, reduced power consumption, and smaller silicon area

These enhancements can transform the project from an educational processor into a robust embedded processing platform suitable for industrial and research applications.

THE ASIC COMPLETE FLOW



INCLUDE THE PHOTOGRAPHS OF THE TEAM HERE